LMapper

CES 2027

Find the leak without scanning the network

SOFTWARE

Ranks the network, issues the coordinates



Detects the district from flow and pressure the utility already records

Ranks every pipe a quasi-2D flow engine scores the whole network

Issues the route exact coordinates, ordered, sent to the crew

HARDWARE

Captures at the coordinate



Handheld scanner
walked to each coordinate



Autonomous survey unit
drives the same route unattended

6.7× **less field time**
same district, same 0.7 m spacing

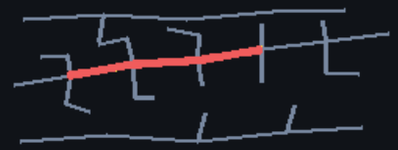
95% **acoustic accuracy**
KTL-witnessed testing

0 **new sensors**
nothing installed underground

Where LMapper changes the method

LMapper

Leak work has always run area → line → point. LMapper keeps the sequence and replaces the weak step in each.

	AREA — which district is losing water	LINE — which pipes to inspect	POINT — is there a leak here
CONVENTIONAL human judgment at every step	 **Someone notices** A person watches flow charts for a change.	 **Scan everything** Listen along every pipe at 0.7 m intervals.	 **One technician's ear** Results vary by listener and street noise.
LMAPPER measurement and models at every step	 **Measured signature** Pressure falls and flow rises. No new sensors.	 **Ranked segments** Only 15% of the pipe length is walked.	 **A model, not an ear** The same sound gets the same answer.
	0 new sensors installed	**6.7×** fewer inspection points	**95%** KTL-verified accuracy

Why two dimensions

LMapper

Ranking a whole district needs speed and detail at once.

1D MODEL

One average velocity



SPEED — fast

DETAIL — coarse

2D · LMAPPER

Resolved across the radius



SPEED — fast

DETAIL — sharp

3D CFD

Every cell, every step



SPEED — slow

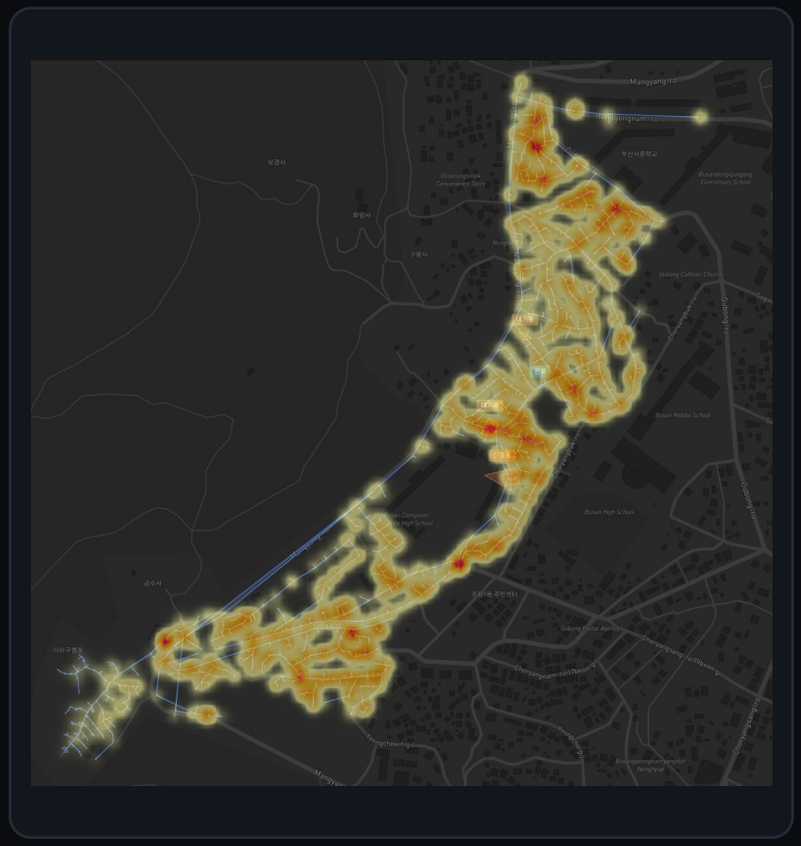
DETAIL — exact

Only the middle one ranks a whole district fast enough to send a crew out the same day.

Built on our founder's published quasi-2D transient work — J. Hydraulic Research 2018, ASME J. Fluids Engineering 2019.

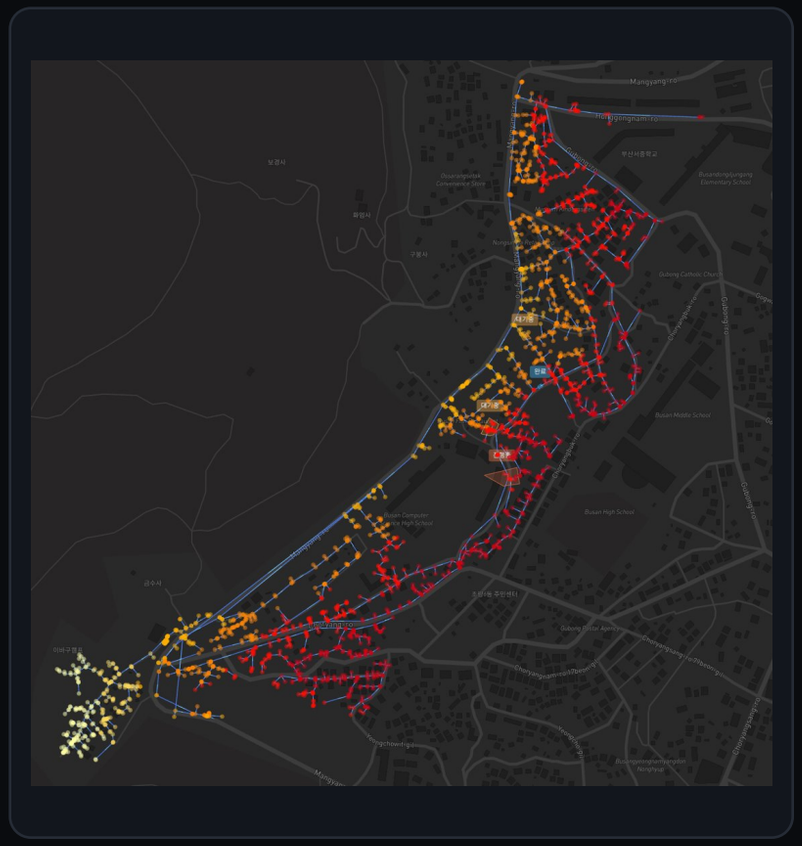
The platform

From district alert to inspection route, in one system.



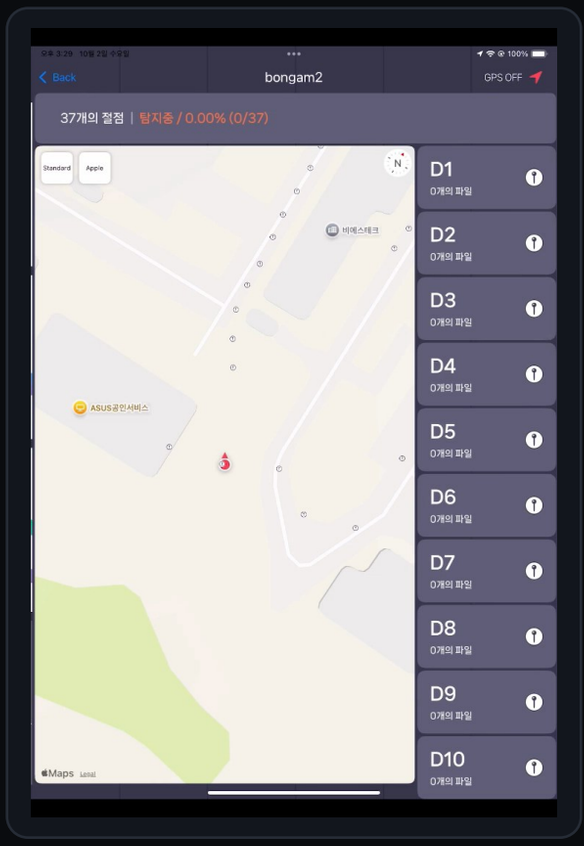
Leak-probability map

Busan pilot district, whole network scored



Ranked inspection nodes

Every 0.7 m point colored by probability



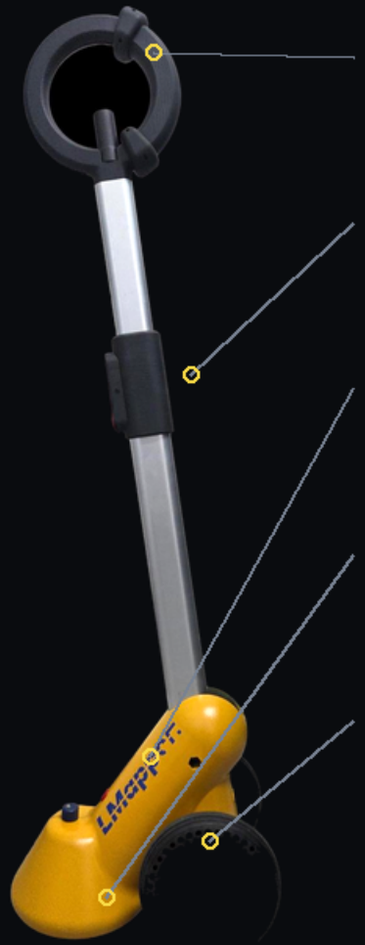
Route on the crew's device

Ordered nodes, one sound file per node

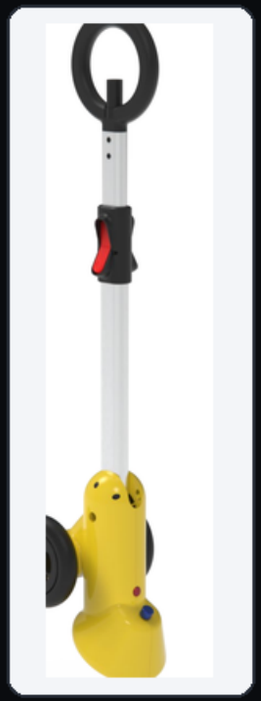
Closed loop. Confirmed leaks carry straight into isolation-area calculation, valve selection, repair scheduling and restoration history.

Handheld scanner

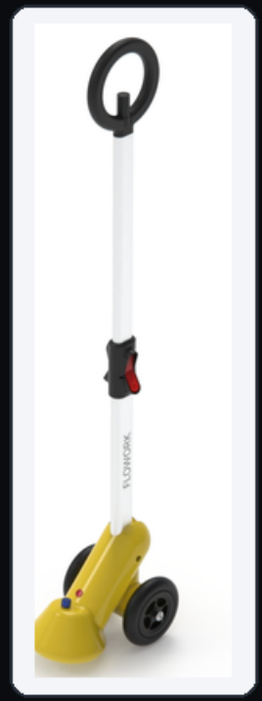
Walked to the coordinate the platform assigns.



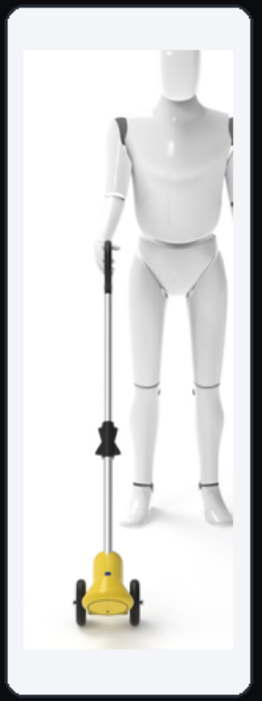
SET POSITIONS AND CARRY



Retracted



Upright



Moving



Tablet case

One grip position, both hands free

The handle slides along the pole, so the hand stays where it started. The tablet case is worn like a messenger bag and holds open at 90°.

Proven in the field

Busan pilot, 2025, and third-party testing.



LMapper Scan on a live survey route

18,766 → 2,792

inspection points

13,136 m network, same 0.7 m spacing

85%

less field time

78 hours cut to 11.6 hours

95%

acoustic accuracy

KTL-witnessed, all 20 simulated leaks found

0

new sensors

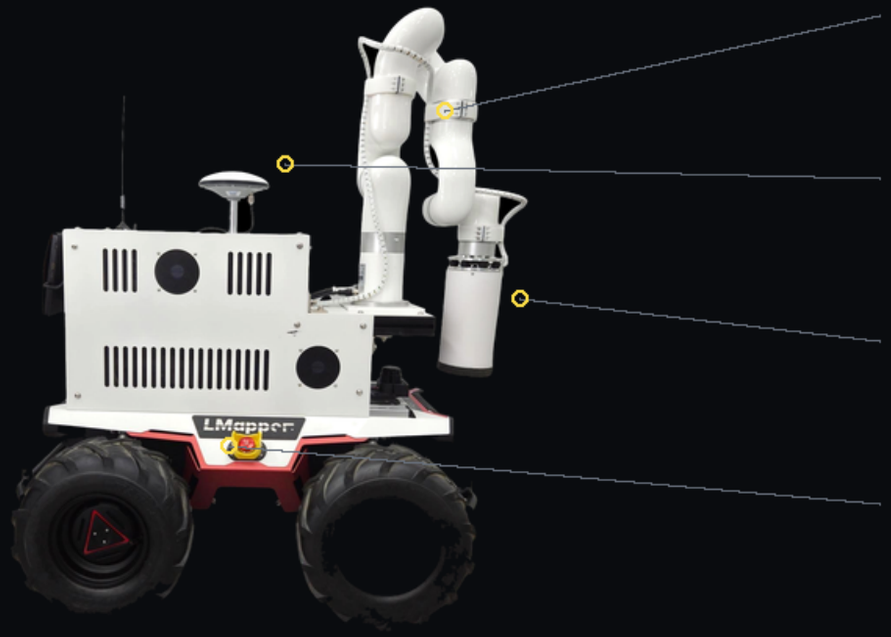
uses data the utility already collects

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11 registered Korean patents, 6 more pending. Digital twin engines and pilots with the cities of Busan, Daegu and Ulsan and with K-water.

Autonomous survey unit

Drives the ranked route and sets the sensor down by itself.



Five-axis arm
600 mm reach · ±0.1 mm repeatability

Positioning
2D lidar · IMU · RTK-GPS

Sensor head
custom housing on rubber dampers

Chassis
independent four-wheel drive

Contact by pressure, not by guess
A pressure sensor between the dampers stops the arm; recording starts right then.

ARM SEQUENCE

1. Approach
2. Ready
3. Lower
4. Press
5. Raise
6. Return

993 × 699 × 349 mm	68 kg	24 V 30 Ah	4 × 400 W	1.5 m/s	30° climb

At Eureka Park

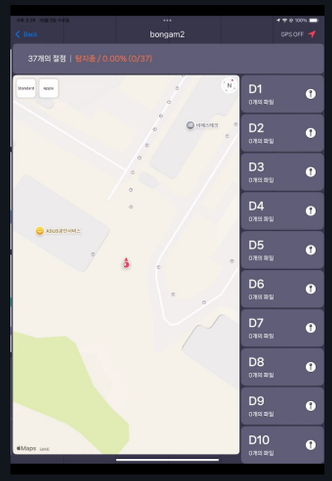
LMapper

Three things on the table, one leak hunt from start to finish.



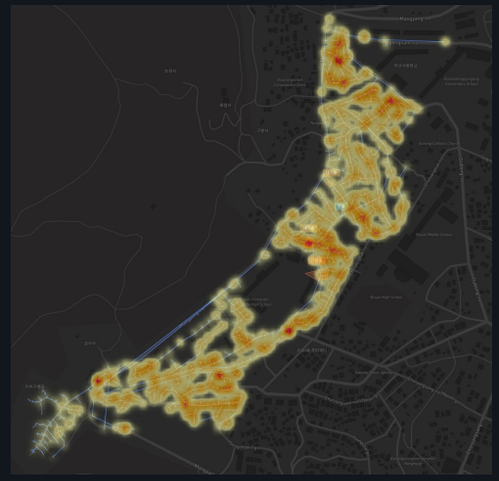
1 LMapper Scan

The handheld, live. Visitors push it to a marked point and set it down, exactly as a field crew does.



2 Field tablet

The crew app: ordered route, node-by-node capture, and the AI's leak call returned on screen.



3 Platform on screen

A real Busan district: alert, pipe ranking, leak-probability map and isolation valves.

LIVE DEMO FLOW

District alert ▶ Pipes ranked ▶ Route to tablet ▶ Scan at the point ▶ AI leak call

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